## Computer Network Course Practice

# Lab-6 Basic OSPF Configuration and Analysis

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## OBJECTIVES

1. Understand static and dynamic routing.
2. Understand OSPF and its fundamental operations.
3. Configure and verify basic OSPF routing.

## REQUIRED RESOURCES

1. Huawei eNSP.
2. ping.
3. tracert.
4. Wireshark.

## BACKGROUND

OSPF (Open Shortest Path First) is an **open standard** interior gateway protocol and **link-state** routing protocol and uses the **Shortest Path First (SPF) algorithm**, developed by Dijkstra, to provide a loop-free topology.
OSPF is a classless protocol and allows for a hierarchical design with VLSM (Variable Length Subnet Masking). Additional OSPF features include equal-cost, multipath routing, and routing based on upper-layer type-of-service (TOS) requests.
OSPF version 2 is used for IPv4. OSPF Version 3 as defined in RFC 2740 is designed for IPv6. Unless otherwise stated, OSPF refers to OSPFv2.

### Area

An OSPF internetwork is subdivided into areas. An OSPF area is a logical group of routers and networks. OSPF implements a **two-layer hierarchy**: the **backbone (Area 0)** and areas off of the backbone (Areas 1–65,535). All areas should connect to Area 0 and all routers will maintain a same topology map of their configured area. All traffic destined between areas, also known as inter-area traffic must traverse Area 0 which is called the backbone area. All area's must have a single interface attaching the area to Area 0.

### Area Types

OSPF areas include common areas, stub areas, and not-so-stubby areas (NSSAs). The brief descriptions of these types are shown in Table 6-1.

Table 6-1 OSPF Area Types

| Area Types | Functions | Descriptions |
|---|---|---|
| Common area | By default, OSPF areas are common areas. Common areas include:<br>**Standard area:** transmits intra-area, inter-area, and external routes.<br>**Backbone area:** connects to all other OSPF areas and transmits inter-area routes. | The backbone area must have all its devices connected.<br>All non-backbone areas must remain connected to the backbone area. |
| Stub area | A stub area is a non-backbone area with only one ABR and generally resides at the border of an AS. The ABR in a stub area does not transmit received AS external routes, which significantly decreases the number of entries in the routing table on the ABR and the amount of routing information to be transmitted. To ensure the reachability of AS external routes, the ABR generates a default route and advertises the route to non-ABRs in the stub area.<br>A totally stub area allows only intra-area routes and ABR-advertised Type 3 LSAs carrying a default route to be advertised within the area. | The backbone area cannot be configured as a stub area.<br>An ASBR cannot exist in a stub area. Therefore, AS external routes cannot be advertised within the stub area.<br>A virtual link cannot pass through a stub area. |
| NSSA | An NSSA is similar to a stub area. An NSSA does not advertise Type 5 LSAs. but can import AS external routes. ASBRs in an NSSA generate Type 7 LSAs to carry the information about the AS external routes. The Type 7 LSAs are advertised only within the NSSA. When the Type 7 LSAs reach an ABR in the NSSA, the ABR translates the Type 7 LSAs into Type 5 LSAs and floods them to the entire AS.<br>A totally NSSA area allows only intra-area routes to be advertised within the area. | ABRs in an NSSA advertise Type 7 LSAs carrying a default route within the NSSA. All inter-area routes must be advertised by ABRs.<br>A virtual link cannot pass through an NSSA. |

### Router Types

Five different types of routers designate the hierarchical routing structure used by OSPF. Each type of routers has a unique role and set of defining characteristics within the hierarchy, as described in Table 6-2.

Table 6-2 OSPF Router Types

| Types of OSPF Routers | Descriptions |
|---|---|
| Internal Router | Routers directly connected networks all belong to the same OSPF area. This type of routers have a single link-state database because they belong to only one area. |

| | |
|---|---|
| Backbone Router | Routers whose interfaces connect them only to the backbone area are considered backbone routers (BRs). BRs do not have an interface to the other OSPF areas, because if they did, they would be considered ABRs. |
| Area Border Router (ABR) | Routers located on the border of one or more OSPF areas and that connect those areas to the backbone area are known as ABRs, which are considered to be members of both the OSPF backbone and the attached areas. The ABRs therefore maintain multiple link-state databases (LSDBs) that describe both the backbone topology and the topology of the other areas. |
| Autonomous System Boundary Router (ASBR) | Routers that connected to external AS outside OSPF AS. ASBRs advertise the exchanged external routing information throughout their AS. Every router within an AS knows how to get to each ASBR with its AS. ASBRs run both OSPF and another routing protocol, such as RIP or BGP. |
| Designated Router (DR) and Backup Designated Router (BDR) | A DR is the router interface elected among all routers on a network segment, and BDR is a backup for the DR. Designated Routers are used for reducing network traffic by providing a source for routing updates. DR maintains a complete topology table of the network and sends the updates to the other routers via multicast. |

**LSA Types**

OSPF provides fast convergence with triggered, incremental updates via Link State Advertisements (LSAs). OSPF Router uses a LSDB (link state database) to maintains its Routing tables. LSAs build up the OSPF link state database called LSDB. All the routers in an OSPF domain exchanges LSA's packets which it contains the link state Database information. By default, OSPF floods new LSAs over all interfaces in the same area, except the interface on which the LSA arrives. Some redundancy is desirable, because it ensures robust flooding.

Instead of using one LSA packet OSPF has many different types of LSAs. Each of LSA types advertises different information in OSPF. Totally OSPF has 11 types of LSA, as shown in Table 6-3. Three most common LSAs are Type 1, 2 and 3.

Table 6-3 OSPF LSA types with a brief description

| LSA Number | LSA Name | Descriptions |
|---|---|---|
| Type 1 | Router LSA | Generated by all routers in an area and list the directly connected networks; this specific LSA do not transit the ABR/ASBR into other areas. |
| Type 2 | Network LSA | Generated by the DR on a multi-access network such as Ethernet to identify all routers to that network segment. This LSA type is flooded throughout the local area only. |
| Type 3 | Summary LSA | Generated by the ABR to describe summary routes to neighboring routers outside of the Area such as an Area 1 ABR summarizing the 10.20.0.0/12 network to Area 0. |
| Type 4 | ASBR Summary LSA | Generated by an ABR to describe route(s) to an ASBR to outside neighboring routers. For example, an ABR telling Area 0 that an ASBR is located inside Area 1. LSA's Type 3 and 4 advertise Inter-Area routes. |
| Type 5 | ASBR External LSA | Generated by an ASBR to describe routes towards an external network such as redistributed networks. |
| Type 6 | Group Membership LSA | Multicast LSA, used for multicast applications. |
| Type 7 | NSSA External LSA | Generated by an ASBR in a Not-So-Stubby-Area (NSSA) which describes a route to an external network. These LSA's are sent to the ABR which in turn translates these LSA's to Type 5 to be sent into the Backbone area. |
| Type 8 | External Attributes LSA (OSPFv2) / Link-Local LSA (OSPFv3) | Used to work with BGP. In OSPFv2 (IPv4) , it is used to transfer BGP attributes throughout an OSPF network. With OSPFv3, This type is redefined to carry IPv6 information throughout the OSPF network. |
| Type 9 | Link Scope Opaque (OSPFv2) / Intra Area Prefix LSA (OSPFv3) | Opaque LSA. This is commonly referred to as a link-state opaque LSA. It is used for intra-area routed within OSPFv3. As link-local advertisements, they are not flooded beyond a single link to another device. |
| Type 10 | Area Scope Opaque LSA | Opaque LSA. Advertised inside an OSPF area. Used to extend OSPF functionality, such as to support traffic engineering (TE). |
| Type 11 | (Autonomous System Scope Opaque LSA | Opaque LSA. Advertised within an AS. Do the same job as Type 10 but are not flooded into special area types like Stub areas. |

**Router ID**

Every Router in an OSPF network needs a unique OSPF Router ID. The OSPF Router ID is used to provide a unique identity to the OSPF Router.

OSPF Router ID is an IPv4 address (32-bit) assigned to each router running the OSPF protocol. OSPF Router ID (RID) must be configured unique for each router in the OSPF Network. Router ID will be selected as highest IP address of any of its configured interfaces. OSPF uses the following criteria to select the Router ID:

1. Manual configuration of the router ID on OSPF Router.
2. OSPF Router select highest IP address of a loopback interfaces on a Router.
3. OSPF Router select highest IP address of a Physical interfaces on a Router.

**OSPF Packets**

OSPF packets are encapsulated into IP packets and use the protocol number 89. OSPF uses the following packet types:

- **Hello:** Periodically sent to find and maintain neighbors, containing the values of some timers, and information about the DR, BDR, and known neighbors.
- **Database description (DD):** Describes the digest of each LSA in the LSDB, exchanged between two routers for data synchronization.
- **Link state request (LSR):** Requests needed LSAs from the neighbor. After exchanging the DD packets, the two routers know which LSAs of the neighbor are missing from their LSDBs. They then send an LSR packet to each other, requesting the missing LSAs.
- **Link state update (LSU):** Transmits the requested LSAs to the neighbor.
- **Link state acknowledgment (LSAck):** Acknowledges received LSU packets.

OSPF uses the destination addresses 225.0.0.5 for all OSPF routers and 224.0.0.6 for communication between the DR/BDR (Designated Router and Backup Designated Router).

# LAB-6-1: BASIC MULTIPLE AREA OSPF CONFIGURATION

## Requirements

The network topology is shown in Figure 6-1. Four routers are used to interconnect the networks of different departments on the campus. Please configure multiple-area OSPF on the four routers to implement communication between departments A, B and loopback 0 (where loopback 0 simulates a host on the network of other departments). The entire AS is divided into three areas.

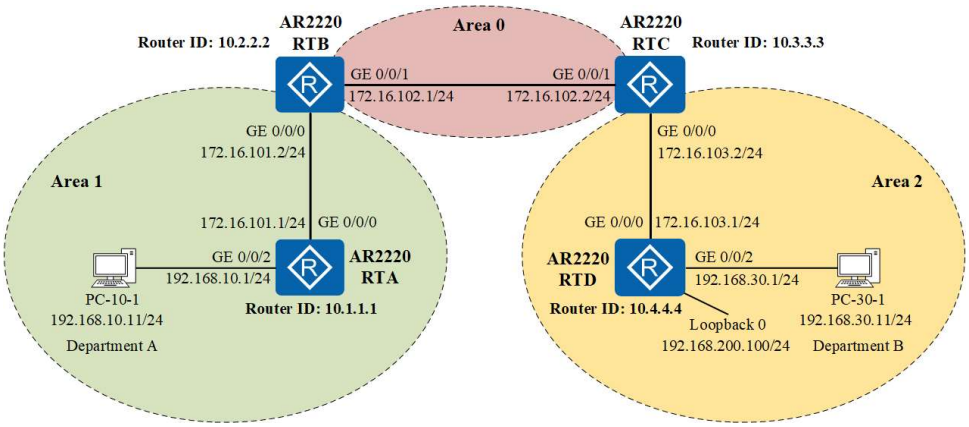


Figure 6-1 A network with OSPF configured on the routers

The IPv4 address allocations are shown in Table 6-4.

Table 6-4 IPv4 Addressing Table

| | IPv4 Address | Mask | Default Gateway |
|---|---|---|---|
| **PC** | | | |
| PC-10-1 | 192.168.10.11 | 255.255.255.0 | 192.168.10.1 |
| PC-30-1 | 192.168.30.11 | 255.255.255.0 | 192.168.30.1 |
| **Router RTA** | | | |
| Router ID | 10.1.1.1 | | |
| GE 0/0/2 | 192.168.10.1 | 255.255.255.0 | |
| GE 0/0/0 | 172.16.101.1 | 255.255.255.0 | |
| **Router RTB** | | | |
| Router ID | 10.2.2.2 | | |
| GE 0/0/0 | 172.16.101.2 | 255.255.255.0 | |
| GE 0/0/1 | 172.16.102.1 | 255.255.255.0 | |
| **Router RTC** | | | |
| Router ID | 10.3.3.3 | | |
| GE 0/0/1 | 172.16.102.2 | 255.255.255.0 | |
| GE 0/0/0 | 172.16.103.2 | 255.255.255.0 | |
| **Router RTD** | | | |
| Router ID | 10.4.4.4 | | |
| GE 0/0/0 | 172.16.103.1 | 255.255.255.0 | |
| GE 0/0/2 | 192.168.30.1 | 255.255.255.0 | |
| Loopback 0 | 192.168.200.100 | 255.255.255.0 | |

## Steps

### Step 1: Load the topology.

1. Start eNSP and click the "Open" icon in the toolbar, and load the topology file **lab-5-1 RT.RIPv1.topo** saved in the Lab-5-1.
2. Check and make sure the settings of the IP address, subnet mask and default gateway of each PC match the address allocation.
3. Click the "Save As" icon in the toolbar to save the topology as **lab-6-1 RT.OSPFv2.MArea.topo**.

### Step 2: Start devices.

Click the " Start Device" icon in the toolbar to start all devices.

**Step 3: Assign IP addresses to interfaces on the routers.**

Assign and check IP addresses to interfaces on each router following the instructions in Step 3 in the Lab-5-1.

**Step 4: Start data captures.**

Start data capture on interface GE 0/0/0 and GE 0/0/1 on router RTB and interface GE 0/0/0 and GE 0/0/1 on router RTC.

**Step 5: Configure basic multiple-area OSPF on the routers.**

1. Configure RTA

Double-click the icon of router RTA in the work area to open its configuration window.

```
<RTA> system-view
# Set the router ID. The router ID uniquely identifies a device in an AS.
[RTA] router id 10.1.1.1
# Creates and run an OSPF process. The value of process id is an integer that ranges from 1 to 65535. The default value is 1.
[RTA] ospf 1
# Creates OSPF area 1.
[RTA-ospf-1] area 1
# Specifies the interface that runs OSPF and the area to which the interface belongs.
# The wildcard mask of an IP address is the inverse of the mask of the IP address.
[RTA-ospf-1-area-0.0.0.1] network 192.168.10.0 0.0.0.255
[RTA-ospf-1-area-0.0.0.1] network 172.16.101.0 0.0.0.255
[RTA-ospf-1-area-0.0.0.1] quit
[RTA-ospf-1] quit
```

2. Configure RTB

```
<RTB> system-view
[RTB] router id 10.2.2.2
[RTB] ospf 1
# Creates OSPF backbone Area 0.
[RTB-ospf-1] area 0
[RTB-ospf-1-area-0.0.0.0] network 172.16.102.0 0.0.0.255
[RTB-ospf-1-area-0.0.0.0] quit
# Creates OSPF area 1.
[RTB-ospf-1] area 1
[RTB-ospf-1-area-0.0.0.1] network 172.16.101.0 0.0.0.255
[RTB-ospf-1-area-0.0.0.1] quit
[RTB-ospf-1] quit
```

3. Configure RTC

```
<RTC> system-view
[RTC] router id 10.3.3.3
[RTC] ospf 1
# Creates OSPF backbone Area 0.
[RTC-ospf-1] area 0
[RTC-ospf-1-area-0.0.0.0] network 172.16.102.0 0.0.0.255
[RTC-ospf-1-area-0.0.0.0] quit
# Creates OSPF area 2.
[RTC-ospf-1] area 2
[RTC-ospf-1-area-0.0.0.2] network 172.16.103.0 0.0.0.255
[RTC-ospf-1-area-0.0.0.2] quit
[RTC-ospf-1] quit
```

4. Configure RTD

```
<RTD> system-view
[RTD] router id 10.4.4.4
```

```
[RTD] ospf 1
# Creates OSPF area 2.
[RTD-ospf-1] area 2
[RTD-ospf-1-area-0.0.0.2] network 172.16.103.0 0.0.0.255
[RTD-ospf-1-area-0.0.0.2] network 192.168.30.0 0.0.0.255
[RTD-ospf-1-area-0.0.0.2] network 192.168.200.0 0.0.0.255
[RTD-ospf-1-area-0.0.0.2] quit
[RTD-ospf-1] quit
```

**Step 6: Examine and verify OSPF configurations.**

1. Examine the OSPF configurations on RTA. You can use the same commands to examine the OSPF configurations on RTB, RTC and RTD.

Double-click the icon of router RTA in the work area to open its configuration window.

```
#   Display OSPF brief information.
<RTA> display ospf brief
<RTA> display ospf 1 brief
# Display information about OSPF interfaces.
<RTA> display ospf interface all
<RTA> display ospf 1 interface gigabitethernet 0/0/0
# Display OSPF Link-State Database (LSDB).
<RTA> display ospf lsdb
<RTA> display ospf 1 lsdb
<RTA> display ospf lsdb brief
<RTA> display ospf 1 lsdb brief
# Display information about router LSAs in the LSDB.
<RTA> display ospf lsdb router
<RTA> display ospf 1 lsdb router
# Display information about network LSAs in the LSDB.
<RTA> display ospf lsdb network 192.168.10.1
<RTA> display ospf 1 lsdb network 172.16.101.0
# Display information about the OSPF neighbor.
<RTA> display ospf peer
<RTA> display ospf 1 peer brief
# Display the OSPF routing table.
<RTA> display ospf routing
<RTA> display ospf 1 routing router-id 10.1.1.1
<RTA> display ospf 1 routing router-id 10.2.2.2
<RTA> display ospf 1 routing router-id 10.3.3.3
<RTA> display ospf 1 routing router-id 10.4.4.4
# Display brief information about route calculation statistics in OSPF processes.
<RTA> display ospf spf-statistics
<RTA> display ospf 1 spf-statistics
<RTA> display ospf 1 spf-statistics verbose
# Display the frequent updates of the LSAs that the LSDB receives.
<RTA> display ospf statistics updated-lsa
<RTA> display ospf 1 statistics updated-lsa
<RTA> display ospf statistics updated-lsa originate-router 10.1.1.1
<RTA> display ospf statistics updated-lsa originate-router 10.2.2.2
<RTA> display ospf statistics updated-lsa originate-router 10.3.3.3
<RTA> display ospf statistics updated-lsa originate-router 10.4.4.4
# Display the update history of LSAs.
<RTA> display ospf statistics updated-lsa history
# display IP routing table.
<RTA> display IP routing-table
# display all OSPF routes in IP routing table.
<RTA> display IP routing-table protocol ospf
```

2. Verify OSPF routing.

(1) Test connectivity between devices.
Ping PC-30-1 from PC-10-1 command window and visa versa.
Ping PC-10-1 from loopback 0 on router RTD configuration window and visa versa.

(2) Trace the route from PC-10-1 to PC-30-1. Issue the following command in PC-10-1's command window:

```
tracert 192.168.30.11
```

**Answer the following questions:**

Q6-1.1. Paste the screenshot of the created topology.

Q6-1.2. Paste the screenshots of the OSPF routing table, Link-State Database, OSPF neighbor, updates of the LSAs that the LSDB receives and route calculation statistics on the router RTA.

Q6-1.3. Paste the screenshots of the OSPF routing table, Link-State Database, OSPF neighbor, updates of the LSAs that the LSDB receives and route calculation statistics on the router RTB.

Q6-1.4. Paste the screenshots of the OSPF routing table, Link-State Database, OSPF neighbor, updates of the LSAs that the LSDB receives and route calculation statistics on the router RTC.

Q6-1.5. Paste the screenshots of the OSPF routing table, Link-State Database, OSPF neighbor, updates of the LSAs that the LSDB receives and route calculation statistics on the router RTD.

Q6-1.6. Can PC-10-1 and PC-30-1 ping each other? Please paste the screenshot of the ping result.

Q6-1.7. Can PC-10-1 and loopback 0 on RTD ping each other? Please paste the screenshot of the ping result.

Q6-1.8. Please paste the screenshot of the result of command "tracert 192.168.30.11" issued from PC-10-1.

Q6-1.9. Just after configuring OSPF on RTA and RTB, what OSPF packets are exchanged between RTA and RTB? Please explain with the captured OSPF communications.

Q6-1.10. With the completion of the OSPF configurations on each router, how does RTA know that the network topology has changed? Please explain with the captured OSPF communications.

Q6-1.11. What OSPF packets are sent and received from the interface GE 0/0/0 and GE 0/0/1 on RTB? Paste the screenshot of the captured OSPF packets.

Q6-1.12. What OSPF packets are sent and received from the interface GE 0/0/0 and GE 0/0/1 on RTC? Paste the screenshot of the captured OSPF packets.

Q6-1.13. In what type of packet are the OSPF Link State Advertisements sent from the interface GE 0/0/0 on RTB transmitted? What are their LSA types? Paste the screenshot of the expanded information of the captured LSAs.

Q6-1.14. In what type of packet are the OSPF Link State Advertisements sent from the interface GE 0/0/1 on RTB transmitted? What are their LSA types? Paste the screenshot of the expanded information of the captured LSAs.

Q6-1.15. How are the Link State Advertisements in area 1 transmitted to area 2 through area 0? Please explain with the captured OSPF communications.

Q6-1.16. How are the Link State Advertisements in area 2 transmitted to area 1 through area 0? Please explain with the captured OSPF communications.

## COMMENTS

Your comments and/or suggestions on this lab: